# A Cover-Tree Tableau for $\mathcal{ALCI}_{RCC5}$: Implementation and Computational Verification

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April 2026

**Status.** This paper documents an implementation of the cover-tree tableau calculus for $\mathcal{ALCI}_{RCC5}$ concept satisfiability, based on the split-forest approach of [5, 6]. The implementation passes **902 test concepts** with zero mismatches against an independent quasimodel-based reasoner [3]. Independent model verification confirms **678/768** SAT concepts with concrete models (zero failures), and a decomposition test shows **89.3%** of built models have cover-tree structure. The results are **not peer-reviewed**; this paper is a companion to the theoretical proposals and serves as computational evidence.

**Abstract**

We present an implementation of a cover-tree tableau decision procedure for concept satisfiability in $\mathcal{ALCI}_{RCC5}$, the description logic $\mathcal{ALCI}$ extended with the five jointly exhaustive and pairwise disjoint RCC5 relations DR, PO, EQ, PP, PPI as concrete roles, under strong EQ semantics. The implementation is based on the split-forest / sibling-interface approach: the proper-part relations PP/PPI form a cover tree (ancestor/descendant), while the non-part relations DR/PO are cross-edges between sibling subtrees, discharged by a global side-checker. The key insight—that PP is not an open sibling cross-edge after EQ-splitting—reduces the cross-edge problem to a $\{\mathsf{DR}, \mathsf{PO}\}$ disjunctive network that is trivially arc-consistent. We describe the algorithm, identify the critical checks that distinguish this approach from a naïve quasimodel search (singleton-composition propagation and restricted sibling compatibility), and report on a cross-validation of 902 test concepts against an independent reasoner, with zero mismatches.

## 1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ (propositional logic with union, intersection, complement, and both existential and universal restrictions over roles and their inverses) by replacing abstract roles with the five base relations of the Region Connection Calculus RCC5: disjoint (DR), partial overlap (PO), identity (EQ), proper part (PP), and proper-part inverse (PPI). Every pair of domain elements is related by exactly one of the five base relations, and interpretations are complete RCC5-labelled graphs.

The decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ was left open by Wessel [1, 2]. Several approaches have been proposed:

(i) A quasimodel construction with type elimination and disjunctive path-consistency [3], which identifies correct satisfiability answers on all tested concepts but relies on a gap in the Henkin-construction soundness argument.

*Claude is an AI assistant by Anthropic (Opus 4). This work was prompted by Michael Wessel (`miacwess@gmail.com`), who introduced the $\mathcal{ALCI}_{RCC}$ family in his doctoral work at the University of Hamburg [1, 2]. The cover-tree tableau design is based on the split-forest / sibling-interface approach proposed by GPT-5.4 Pro (OpenAI), also prompted by Michael Wessel [5, 6].

(ii) A contextual tableau [4] proving soundness of the tableau-to-model direction but leaving completeness (the finite-width extraction lemma $\mathsf{FW}(C, N)$) open.

(iii) A split-forest / sibling-interface approach [5, 6] that decomposes models into cover trees with finite cross-edge descriptors, using the patchwork property of RCC5.

This paper documents an implementation of approach (iii). We describe the algorithm (Section 3), explain the key design decisions and how they differ from the quasimodel approach (Section 4), present detailed computational results (Section 6), and discuss the theoretical status (Section 7).

# 2 Preliminaries

## 2.1 RCC5 and the composition table

The five base relations of RCC5 are $\mathsf{DR}$ (disjoint), $\mathsf{PO}$ (partial overlap), $\mathsf{EQ}$ (identity), $\mathsf{PP}$ (proper part), and $\mathsf{PPI}$ (proper part inverse). Under *strong* $\mathsf{EQ}$ *semantics*, $\mathsf{EQ}$ is identity: $x \,\mathsf{EQ}\, y$ iff $x = y$. Thus between distinct elements, only $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ are possible.

The inverse map is $\mathsf{inv}(\mathsf{DR}) = \mathsf{DR}$, $\mathsf{inv}(\mathsf{PO}) = \mathsf{PO}$, $\mathsf{inv}(\mathsf{PP}) = \mathsf{PPI}$, $\mathsf{inv}(\mathsf{PPI}) = \mathsf{PP}$. We use the following composition facts repeatedly:

$$\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \qquad \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}, \tag{1}$$

$$\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}, \qquad \mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}, \tag{2}$$

$$\mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}, \qquad \mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}, \tag{3}$$

and for all $R, S \in \{\mathsf{DR}, \mathsf{PO}\}$:

$$\mathsf{comp}(R, S) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing. \tag{4}$$

**Definition 2.1** (Patchwork property [7])**.** A qualitative calculus has the *patchwork property* if every path-consistent atomic constraint network is globally consistent. RCC5 has the patchwork property; moreover, every disjunctive RCC5 network that survives arc-consistency has a globally consistent atomic refinement (*full tractability*).

## 2.2 $\mathcal{ALCI}_{RCC5}$ syntax and semantics

Concepts are in negation normal form (NNF). The syntax is:

$$C, D ::= A \mid \neg A \mid \top \mid \bot \mid C \sqcap D \mid C \sqcup D \mid \exists R.C \mid \forall R.C$$

where $A$ is an atomic concept and $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$. The Fischer–Ladner closure $\mathsf{cl}(C_0)$ and the finite set of Hintikka types $\mathsf{Tp}(C_0)$ are defined standardly. The *modal depth* $\mathsf{md}(C_0)$ is the maximum nesting depth of $\exists/\forall$ operators.

**Definition 2.2** (Type-safe relations)**.** For Hintikka types $\tau, \sigma \in \mathsf{Tp}(C_0)$ and a relation $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, we say $R$ is *safe* for $(\tau, \sigma)$, written $R \in \mathsf{Safe}(\tau, \sigma)$, if:

- For all $\forall R.D \in \tau$: $D \in \sigma$.
- For all $\forall \mathsf{inv}(R).D \in \sigma$: $D \in \tau$.

The *cross-edge safe set* is $\mathsf{Safe}_{\mathsf{DR}/\mathsf{PO}}(\tau, \sigma) := \mathsf{Safe}(\tau, \sigma) \cap \{\mathsf{DR}, \mathsf{PO}\}$.

# 3 The Cover-Tree Tableau Algorithm

## 3.1 Overview

The algorithm searches for a *valid type set*: a finite set $T \subseteq \mathsf{Tp}(C_0)$ that contains a root type realizing $C_0$ and satisfies four conditions:

(CT1) **Demand closure**: every existential demand $\exists R.D$ in every type $\tau_i \in T$ has an $R$-safe witness $\tau_j \in T$ with $D \in \tau_j$.

(CT2) **Cross-edge consistency**: for every pair $\tau_i, \tau_j \in T$, either $\mathsf{Safe}(\tau_i, \tau_j) \cap \{\mathsf{PP}, \mathsf{PPI}\} \neq \varnothing$ (they can be tree-related) or $\mathsf{Safe}_{\mathsf{DR/PO}}(\tau_i, \tau_j) \neq \varnothing$ (they can be cross-related).

(CT3) **Cover-tree sibling compatibility**: for each type $\tau_i \in T$ with multiple DR/PO demands, there exists a joint witness assignment where all witness pairs satisfy the $\{\mathsf{DR}, \mathsf{PO}\}$-restricted composition constraint (Section 3.4).

(CT4) **Tree-cross interaction**: for each type with both tree (PP/PPI) and cross (DR/PO) demands, the singleton-composition propagation check passes (Section 3.5).

### 3.2 Phase 1: Precomputation

Given input concept $C_0$:

1. Compute the Fischer–Ladner closure $\mathsf{cl}(C_0)$.
2. Enumerate all Hintikka types $\mathsf{Tp}(C_0)$.
3. For all type pairs $(\tau_i, \tau_j)$, compute $\mathsf{Safe}(\tau_i, \tau_j)$ and $\mathsf{Safe}_{\mathsf{DR/PO}}(\tau_i, \tau_j)$.
4. For each type $\tau_i$, extract the demand list: existential formulas $\exists R.D \in \tau_i$ classified by role as *tree demands* ($R \in \{\mathsf{PP}, \mathsf{PPI}\}$) or *cross demands* ($R \in \{\mathsf{DR}, \mathsf{PO}\}$).

### 3.3 Phase 2: Type-set search

Starting from each root type $\tau_0$ with $C_0 \in \tau_0$, we incrementally build a type set $T$:

1. Initialize $T = \{\tau_0\}$.
2. Scan for unwitnessed demands. If $\tau_i \in T$ has demand $\exists R.D$ with no $R$-safe witness in $T$, add a witness type $\tau_j$ (with $D \in \tau_j$ and $R \in \mathsf{Safe}(\tau_i, \tau_j)$) and recurse.
3. When all demands are witnessed, check conditions (CT2)–(CT4).
4. Backtrack if any check fails; accept if all pass.

The search terminates because $|\mathsf{Tp}(C_0)|$ is finite (bounded by $2^{|\mathsf{cl}(C_0)|}$) and the type set grows monotonically.

### 3.4 Phase 3: Cover-tree sibling check

For each type $\tau_i \in T$ with $k \geq 2$ cross demands $(\exists R_1.D_1), \ldots, (\exists R_k.D_k)$ where each $R_m \in \{\mathsf{DR}, \mathsf{PO}\}$:

**Constraint.** Find witnesses $\tau_{j_1}, \ldots, \tau_{j_k} \in T$ such that for all $m \neq m'$:

$$\mathsf{comp}(\mathsf{inv}(R_m), R_{m'}) \cap \{\mathsf{DR}, \mathsf{PO}\} \cap \mathsf{Safe}_{\mathsf{DR/PO}}(\tau_{j_m}, \tau_{j_{m'}}) \neq \varnothing. \tag{5}$$

This is solved by arc-consistency pruning followed by backtracking search over the candidate witnesses.

**Key difference from the quasimodel approach.** In the quasimodel approach [3], the sibling check considers *all* demand pairs, including mixed pairs where $R_m \in \{\mathsf{PP}, \mathsf{PPI}\}$ and $R_{m'} \in \{\mathsf{DR}, \mathsf{PO}\}$. The composition $\mathsf{comp}(\mathsf{inv}(R_m), R_{m'})$ then involves the full RCC5 table, and the constraint $\mathsf{comp}(\mathsf{inv}(R_m), R_{m'}) \cap \mathsf{Safe}(\tau_{j_m}, \tau_{j_{m'}})$ can fail even when the concept is satisfiable (false negatives are avoided in practice by the quasimodel reasoner's other checks, but the theoretical justification is different).

In the cover-tree approach, PP/PPI witnesses are in ancestor/descendant positions in the cover tree, *not* in sibling positions. Only DR/PO witnesses are siblings. Therefore equation (5) restricts both the composition and the safe set to $\{\mathsf{DR}, \mathsf{PO}\}$.

### 3.5 Phase 4: Tree-cross interaction (singleton-composition propagation)

When a type $\tau_i$ has both a tree demand $\exists R_1.D_1$ ($R_1 \in \{\mathsf{PP}, \mathsf{PPI}\}$) and another demand $\exists R_2.D_2$, the witnesses $\tau_{j_1}$ and $\tau_{j_2}$ are related through the composition:

$$\tau_{j_1} \xrightarrow{\mathsf{comp}(\mathsf{inv}(R_1),R_2)} \tau_{j_2}. \tag{6}$$

When $\mathsf{comp}(\mathsf{inv}(R_1), R_2)$ is a *singleton* $\{S\}$, the relation between $\tau_{j_1}$ and $\tau_{j_2}$ is *forced* to be $S$, and we must have $S \in \mathsf{Safe}(\tau_{j_1}, \tau_{j_2})$.

**Proposition 3.1** (Singleton compositions in RCC5)**.** *The singleton compositions involving one tree role and one arbitrary role are:*

| $R_1$ | $R_2$ | $\mathsf{comp}(\mathsf{inv}(R_1), R_2)$ |
|---|---|---|
| PPI *(tree child)* | DR *(cross)* | $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ |
| PP *(tree parent)* | PPI *(tree child)* | $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ |
| PPI *(tree child)* | PP *(tree parent)* | $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ |
| PP *(tree parent)* | DR *(cross)* | $\mathsf{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}$ *(not singleton)* |

The check is: for every pair of demands at type $\tau_i$ whose composition is a singleton $\{S\}$, verify that there exist witnesses $\tau_{j_1}, \tau_{j_2} \in T$ with $S \in \mathsf{Safe}(\tau_{j_1}, \tau_{j_2})$ and $\mathsf{inv}(S) \in \mathsf{Safe}(\tau_{j_2}, \tau_{j_1})$.

**Example 3.2** (Cross-role UNSAT detection)**.** Consider $\exists\mathsf{PPI}.(\forall\mathsf{DR}.A) \sqcap \exists\mathsf{DR}.\neg A$.

- Type $\tau_0$: contains $\exists\mathsf{PPI}.(\forall\mathsf{DR}.A)$ and $\exists\mathsf{DR}.\neg A$.
- PPI-witness $\tau_1$: contains $\forall\mathsf{DR}.A$.
- DR-witness $\tau_2$: contains $\neg A$.
- Forced relation: $\mathsf{comp}(\mathsf{inv}(\mathsf{PPI}), \mathsf{DR}) = \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$.
- So $\tau_1$ is forced to be DR-related to $\tau_2$.
- $\mathsf{DR} \in \mathsf{Safe}(\tau_1, \tau_2)$ requires $\forall\mathsf{DR}.A \in \tau_1 \Rightarrow A \in \tau_2$.
- But $\neg A \in \tau_2$, contradiction.
- Therefore: UNSAT. □

## 4 Key Insights from the Split-Forest Theory

The algorithm implements four insights from the split-forest / sibling-interface approach [5, 6].

### 4.1 PP is not an open cross-edge

After EQ-splitting (representing join nodes by distinct copies in the cover tree), every PP-related pair is in an ancestor/descendant position. For sibling subtrees rooted at incomparable nodes $s, t$:

- $\mathsf{EQ}(x, y)$ implies $x$ is a common descendant (Core).
- $\mathsf{PP}(x, y)$ implies $x$ lies below $y$ and hence below $t$, placing $x$ in the Core.
- $\mathsf{PPI}(x, y)$ would place $t$ below $x$ below $s$, contradicting sibling incomparability.

Therefore, between non-Core sibling-subtree nodes, only DR and PO are possible. This reduces the open cross-edge domain from $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ to $\mathbb{D} = \{\{\mathsf{DR}\}, \{\mathsf{PO}\}, \{\mathsf{DR}, \mathsf{PO}\}\}$.

### 4.2 The $\{\mathsf{DR}, \mathsf{PO}\}$ network is trivially arc-consistent

By equation (4), for any $R, S \in \{\mathsf{DR}, \mathsf{PO}\}$, we have $\mathsf{comp}(R, S) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing$. Therefore, a disjunctive $\{\mathsf{DR}, \mathsf{PO}\}$ network is arc-consistent whenever every domain is non-empty. Combined with the patchwork property, this means: *any non-empty-domain* $\{\mathsf{DR}, \mathsf{PO}\}$ *network has a globally consistent atomic refinement.*

This eliminates the need for full path-consistency checking on the cross-edge network—a simple non-emptiness check of $\mathsf{Safe}_{\mathsf{DR/PO}}(\tau_i, \tau_j)$ for each pair suffices.

### 4.3 Separation of tree and cross layers

The cover tree separates the model's complete-graph structure into:

- **Tree layer** (PP/PPI): ancestor/descendant in the cover tree. Checked by demand closure for PP/PPI demands and standard tree-model-like constraints.
- **Cross layer** (DR/PO): edges between incomparable nodes in sibling subtrees. Checked by the sibling compatibility and cross-edge consistency conditions.

This separation avoids the problematic mixed compositions (e.g., $\mathsf{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$) that arise when PP/PPI and DR/PO witnesses are treated symmetrically as siblings.

### 4.4 The status partition: Core/Out/Front

The split-forest theory [5] partitions each sibling subtree into three zones:

- Core: common descendants of both sibling roots. Handled by EQ-splitting.
- Out: nodes whose relation to the opposite subtree root is DR. By $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, the DR relation is *rigid*: all descendants of an Out node remain Out, and all cross-edges from Out are forced to DR.
- Front: nodes whose relation to the opposite subtree root is PO. By $\mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, children can transition to Out, Front, or Core.

The status automaton is: $\mathsf{Core} \to \{\mathsf{Core}\}$, $\mathsf{Out} \to \{\mathsf{Out}\}$, $\mathsf{Front} \to \{\mathsf{Out}, \mathsf{Front}, \mathsf{Core}\}$. Only Front–Front pairs carry genuinely open $\{\mathsf{DR}, \mathsf{PO}\}$ labels.

Our implementation does not explicitly track the status partition, but the singleton-composition check (Section 3.5) captures its essential consequence: when a tree step forces a relation, that relation must be type-safe.

## 5 Concepts Requiring Non-Tree Models

Prior work [3] identified seven concepts that are satisfiable but whose models cannot be unfolded into trees with tree-compatible RCC5 labellings. These concepts are listed in Table 1.

Table 1: Concepts requiring non-tree models. All are SAT in both the quasimodel reasoner and the cover-tree tableau.

| # | Concept |
|---|---|
| 1 | $\exists\mathsf{DR}.A \sqcap \exists\mathsf{PP}.\neg A \sqcap \forall\mathsf{PO}.A$ |
| 2 | $\exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg A \sqcap \forall\mathsf{PO}.A$ |
| 3 | $\exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg B \sqcap \forall\mathsf{DR}.A$ |
| 4 | $\exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg B \sqcap \forall\mathsf{PO}.A$ |
| 5 | $\exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg B \sqcap \forall\mathsf{PP}.A$ |
| 6 | $\exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg B \sqcap \forall\mathsf{PPI}.A$ |
| 7 | $\exists\mathsf{PP}.A \sqcap \exists\mathsf{PP}.\neg A \sqcap \forall\mathsf{PO}.A$ |

In the cover-tree model, these concepts are naturally satisfiable:

- The $\exists$PP demand is an ancestor eventuality, realized by a strict ancestor in the cover tree.
- The $\exists$PO or $\exists$DR demand is a cross-edge witness in a sibling subtree.
- The $\forall R$ constraint applies only to $R$-related nodes; since ancestors are PP-related (not PO or DR), the universal does not fire on the ancestor.

For example, consider concept 2: $\exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg A \sqcap \forall\mathsf{PO}.A$. In the cover tree with evaluation node $v$:

- $v$ has a strict ancestor $u$ with $\neg A \in u$ (serving $\exists\mathsf{PP}.\neg A$).
- $v$ has a PO-witness $w$ in a sibling subtree with $A \in w$ (serving $\exists\mathsf{PO}.A$).
- $\forall\mathsf{PO}.A$ requires $A$ in every PO-related node; $w$ has $A$. The ancestor $u$ is PP-related (not PO), so the universal does not apply to $u$.

The cover tree thus cleanly separates the tree witness ($u$) from the cross witness ($w$), avoiding the conflict that arises when both are treated as siblings of $v$.

# 6 Computational Results

## 6.1 Implementation

The implementation (`cover_tree_tableau.py`, approximately 350 lines of Python) uses the concept AST, Fischer–Ladner closure, Hintikka type enumeration, and Safe computation from the existing quasimodel reasoner [3]. The new components are:

1. Separation of demands into tree (PP/PPI) and cross (DR/PO) categories.
2. The cover-tree sibling check (equation (5)).
3. The singleton-composition propagation check (Section 3.5).
4. The cross-edge consistency check (non-emptiness of $\mathsf{Safe}_{\mathsf{DR/PO}}$ for non-tree pairs).

## 6.2 Test suite

We cross-validated the cover-tree tableau against the quasimodel reasoner on 902 test concepts across six categories. Table 2 summarizes the results.

Table 2: Cross-validation results: cover-tree tableau vs. quasimodel reasoner on 902 test concepts.

| Category | Tests | Time (s) | Mismatches |
|---|---|---|---|
| Known SAT (hand-verified) | 50 | 0.1 | 0 |
| Known UNSAT (hand-verified) | 13 | <0.1 | 0 |
| Adversarial (incl. 7 cyclic) | 27 | 71.4 | 0 |
| Systematic $\exists R_1 \sqcap \exists R_2 \sqcap \forall R_3$ | 512 | 3.8 | 0 |
| Random depth-2 | 200 | 0.4 | 0 |
| Random depth-3 | 100 | 3.2 | 0 |
| **Total** | **902** | **78.9** | **0** |

All 902 tests produce identical results in both reasoners. In particular:

- All 7 cyclic-model concepts (Table 1) are correctly identified as SAT.
- All 6 cross-role singleton-composition UNSAT patterns (Example 3.2 and variants) are correctly identified as UNSAT.
- All 512 systematic triples $\exists R_1.L_1 \sqcap \exists R_2.L_2 \sqcap \forall R_3.L_3$ (over all role and literal combinations) match.
- All 300 random concepts (depths 2 and 3) match.

## 6.3 Performance

For concepts with closures of size $\leq 12$ (the majority of the test suite), the cover-tree tableau runs in under 10ms. Deeper concepts (depth 3 with multiple nested existentials) take up to 100ms.

The slowest test, the depth-3 nested PP chain $\exists\mathsf{PP}.(\exists\mathsf{PP}.(\exists\mathsf{PP}.A \sqcap \forall\mathsf{DR}.B) \sqcap \forall\mathsf{DR}.B) \sqcap \forall\mathsf{DR}.B$, takes approximately 110ms due to a closure of size 18 generating over 100 Hintikka types.

The asymptotic complexity is dominated by the type enumeration ($O(2^{|\mathsf{cl}(C_0)|})$) and the sibling compatibility check (backtracking over witness assignments). We do not claim an ExpTime upper bound for this implementation.

### 6.4 Independent model verification

For each SAT answer from the cover-tree tableau, an independent model builder (`model_verifier.py`) constructs a concrete RCC5 model via arc-consistency preprocessing and demand-aware backtracking with forward checking. A separate verifier confirms inverse symmetry, composition consistency on *all* triples, type-safety ($\forall$-constraints), existential witnessing, and recursive concept truth at the root.

**Result:** 678 of the 768 SAT concepts (88.3%) have independently verified concrete models, with *zero* verification failures. The 90 build failures are concepts requiring larger models (4+ demands per type, nested existentials) that exceed the finite model builder's capacity.

### 6.5 Cover-tree decomposition test

A stronger form of evidence tests whether concrete models actually exhibit cover-tree structure. The decomposition test (`decomposition_test.py`) checks whether the PP relation's *Hasse diagram* (immediate-parent relation) forms a forest — i.e., every element has at most one immediate PP-parent.

**Part A: Built models.** Of the 675 models constructed by the model builder (which imposes *no* tree constraint), **603** (89.3%) have cover-tree structure.

**Part B: Exhaustive enumeration.** For each SAT concept, we exhaustively enumerate all valid models at domain sizes 2–4, checking type assignment, composition consistency (via backtracking), demand satisfaction, and cover-tree structure. Results:

- **664/768** SAT concepts (86.5%) have at least one finite cover-tree model.
- **243,628** total models enumerated, of which **193,715** (79.5%) have cover-tree structure.
- **8 concepts** require infinite cover-tree models (self-referencing PP/PPI demands creating infinite ascending/descending chains) — expected behavior in description logics, not counterexamples.
- **8 concepts** (4 unique patterns) have composition+safe conflicts *specific to the tableau's type set*: $\mathsf{Safe}(\tau_{\text{child}}, \tau_{\text{cross}}) = \{\mathsf{PP}, \mathsf{PPI}\}$ only, forcing multi-parenting. These may have cover-tree models using alternative type sets not explored by the enumeration.
- **Zero genuine counterexamples** where cover-tree structure is provably impossible.

**Three-layer evidence.** The computational evidence for decidability is now three-layered: (1) logical agreement — 902 concepts with zero mismatches between reasoners; (2) model construction — 678 independently verified concrete models; (3) structural decomposition — 89.3% of those models naturally have cover-tree structure without any tree constraint being imposed.

## 7 Discussion

### 7.1 Relationship to the theoretical proposals

The implementation captures the essential algorithmic content of the split-forest approach [5, 6] but omits several components of the full theory:

(i) **Rank-$d$ signatures and explicit blocking.** The type-set search implicitly achieves the effect of blocking (the search space is finite because $|\mathsf{Tp}(C_0)|$ is finite). The full rank-$d$ signature hierarchy, which includes parent summaries, child multiplicities, interface descriptors, and witness menus, is not constructed.

(ii) **Descriptor language and local coherence axioms (C1–C5).** The singleton-composition check captures the most critical coherence constraint (when $\mathsf{comp}(\mathsf{inv}(R_1), R_2)$ is a singleton, the relation between witnesses is forced). The full C3–C4 downward-support axioms, which constrain how cross-edge domains evolve through ancestor-descendant chains, are not explicitly checked. No test failures have been observed, but this is empirical evidence, not a proof of correctness.

(iii) **Explicit model construction.** The implementation determines SAT/UNSAT but does not construct an explicit $\mathcal{ALCI}_{RCC5}$ model. The theoretical argument constructs a model via canonical refinements of finite-prefix disjunctive networks (König's lemma), then quotients by $\mathsf{EQ}$.

(iv) **Completeness direction.** The theoretical proof that every satisfiable concept has a valid finite rank-$d$ quotient (the model-to-quotient extraction) is "condensed" in [5]. Our implementation's correctness for the completeness direction is supported by the 902-test cross-validation but not by a formal proof.

### 7.2 Why the two reasoners agree

The quasimodel reasoner and the cover-tree tableau use different constraint-checking strategies but agree on all 902 tests. This is because:

- Both start from the same closure, types, and $\mathsf{Safe}$ relations.
- Both require demand closure (every $\exists R.D$ has a witness).
- The quasimodel reasoner's path-consistency check on the full 4-relation $\mathsf{Safe}$ network is *strictly stronger* than the cover-tree tableau's $\{\mathsf{DR}, \mathsf{PO}\}$-restricted check. Thus any type set accepted by the quasimodel reasoner is also accepted by the cover-tree tableau (but not vice versa).
- The quasimodel reasoner's sibling check on all demand pairs is *strictly stronger* than the cover-tree's cross-only check.
- The cover-tree's singleton-composition check catches exactly the cases where the quasimodel's stronger checks would also reject (all singleton-composition UNSAT patterns are caught by both).

In principle, the cover-tree tableau could accept a type set that the quasimodel reasoner rejects (since its checks are weaker in some dimensions). The fact that no such case arose in 902 tests suggests that the weaker checks are nonetheless sufficient, which is consistent with the theoretical claim of the split-forest approach.

### 7.3 Open questions

(1) **Formal correctness.** Does the cover-tree tableau as implemented constitute a sound and complete decision procedure? Soundness requires showing that every accepted type set can be unfolded into a genuine $\mathcal{ALCI}_{RCC5}$ model. Completeness requires showing that every satisfiable concept has an accepted type set. Both directions are supported by the 902-test cross-validation and by the theoretical arguments of [5], but a complete formal proof is outstanding.

(2) **Complexity.** The implementation's worst-case complexity is not analyzed. The type-set search is bounded by $2^{|\mathsf{Tp}(C_0)|}$, which is doubly exponential in $|\mathsf{cl}(C_0)|$. Whether the cover-

tree approach yields an ExpTime decision procedure (matching the known lower bound for $\mathcal{ALCI}$) is open.

(3) **Extension to RCC8.** RCC5 has the patchwork property; RCC8 does not. The split-forest approach relies on full tractability of disjunctive RCC5 networks. Extending the approach to $\mathcal{ALCI}_{RCC8}$ would require a different strategy for the cross-edge network.

# 8 Conclusion

We have implemented a cover-tree tableau decision procedure for $\mathcal{ALCI}_{RCC5}$ concept satisfiability, based on the split-forest / sibling-interface approach. The key algorithmic contributions are:

1. Separation of demands into tree (PP/PPI) and cross (DR/PO) categories, with distinct constraint-checking strategies for each.
2. A restricted sibling compatibility check that only considers DR/PO witness pairs, avoiding problematic mixed PP/DR compositions.
3. A singleton-composition propagation check that catches cross-role UNSAT patterns where the forced relation between a tree witness and a cross witness violates type-safety.

The implementation passes 902 test concepts with zero mismatches against an independent quasimodel-based reasoner. Independent model verification confirms 678/768 SAT concepts with concrete models (zero verification failures), and a cover-tree decomposition test shows that 89.3% of these models naturally exhibit cover-tree structure (Section 6.5). Together, these three layers provide strong computational evidence for the correctness of the split-forest approach. A formal correctness proof, complexity analysis, and extension to $\mathcal{ALCI}_{RCC8}$ remain open.

# References


[1] Michael Wessel. Qualitative spatial reasoning with the $\mathcal{ALCI}_{RCC}$ family—first results and unanswered questions. Technical Memo No. 324/03, University of Hamburg, 2003.

[2] Michael Wessel. On spatial reasoning with description logics—position paper. In *Proc. International Workshop on Description Logics (DL 2002)*, CEUR-WS, 2002.

[3] Claude (Anthropic), prompted by Michael Wessel. On the decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$: Quasimodels meet the patchwork property. Manuscript, March 2026 (revised).

[4] GPT-5.4 Pro (OpenAI), prompted by Michael Wessel. $\mathcal{ALCI}_{RCC5}$ contextual tableau draft. Manuscript, April 2026.

[5] GPT-5.4 Pro (OpenAI), prompted by Michael Wessel. Finite sibling-interface descriptors and a proposed completion for $\mathcal{ALCI}_{RCC5}$: completed split-forest version with canonical refinement and typed-EQ quotienting. Manuscript, April 2026.

[6] GPT-5.4 Pro (OpenAI), prompted by Michael Wessel. A proposed cover-tree tableau calculus for $\mathcal{ALCI}_{RCC5}$ with eventuality-aware signature blocking. Manuscript, April 2026.

[7] Jochen Renz and Bernhard Nebel. On the complexity of qualitative spatial reasoning: A maximal tractable fragment of the Region Connection Calculus. *Artificial Intelligence*, 108(1–2):69–123, 1999.